

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2018-2019)

PRE-MEDICAL : NURTURE COURSE [PHASE-I, II & III]

TEST # 13

DATE : 06-01-2019

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
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MINOR TEST # 13

Topics

DATE : 06 - 01 - 2019**PRE-MEDICAL NURTURE COURSE [PHASE-I, II & III] : [NEET]****PHYSICS** : NLM & Friction, Wave Motion**CHEMISTRY** : Chemical Bonding, Thermodynamics, p-Block, IUPAC, Nomenclature**BIOLOGY** : Photosynthesis, Respiration, Excretory System & Muscles**SPACE FOR ROUGH WORK**

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BEWARE OF NEGATIVE MARKING

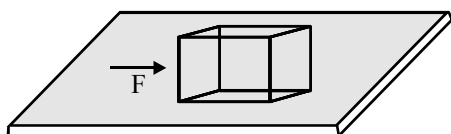
1. A 1 kg block is resting on a vertical wall when a force F is applied perpendicular to it. If coefficient of friction is 0.2 then the minimum value of F is :-

(1) 950 N (2) 49 N
(3) 98 N (4) 490 N

2. A block is resting on an inclined plane. When the inclination is 60° then block starts sliding then coefficient of static friction is :-

(1) 1.173 (2) 1.732
(3) 2.732 (4) 1.677

3. A 2 kg block is resting on horizontal surface. If coefficient of static friction is 0.4 and 2.5 N force is applied on the block then the friction on the block is :-

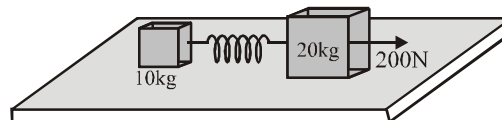


(1) 2.5 N (2) 5N
(3) 7.84 N (4) 10 N

4. A 5 kg is sliding with constant velocity on an inclined plane of inclination 30° then coefficient of friction is :-

(1) $\frac{1}{\sqrt{3}}$ (2) $\frac{2}{\sqrt{3}}$
(3) $\sqrt{3}$ (4) $2\sqrt{3}$

5. In the given figure if the acceleration of 10 kg block is 12 m/s^2 then the acceleration of 20 kg block is :-



(1) 12 m/s^2 (2) 4 m/s^2
(3) 10 m/s^2 (4) Zero

6. A rocket consumes fuel at the rate of 100 kg/s. The exhaust gases are ejected at a speed of $5 \times 10^4 \text{ m/s}$. Neglecting the effect of gravity, the force experienced by the rocket is :-

(1) $5 \times 10^2 \text{ N}$
(2) $5 \times 10^4 \text{ N}$
(3) $5 \times 10^6 \text{ N}$
(4) 50 N

7. A 90 kg man stands in an elevator that has a downward acceleration of 1.4 m/s^2 . The force exerted by him on the floor is about:-

(1) Zero (2) 900 N
(3) 774 N (4) 880 N

8. The Newton's laws of motion are valid in :-

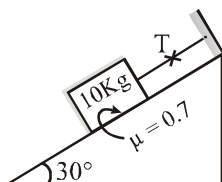
(1) inertial frames
(2) Non-inertial frames
(3) Rotating frames
(4) Accelerated frames

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9. A force $\vec{F} = 6\hat{i} - 8\hat{j} + 10\hat{k}$ newton produces acceleration 1 m/s^2 in a body. The mass of the body is (in kg) :-

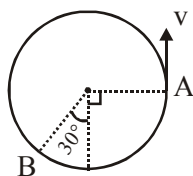
(1) $6\hat{i} - 8\hat{j} + 10\hat{k}$ (2) 100
(3) $10\sqrt{2}$ (4) 10

10. The tension T in the string shown in figure is :-



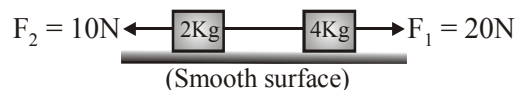
(1) Zero (2) 50 N
(3) $35\sqrt{3}$ N (4) $(\sqrt{3} - 1)50$ N

11. A particle of mass m moves with constant speed v on a circular path of radius r as shown in figure. The average force on it during its motion from A to B is :-



(1) $\frac{\sqrt{3}mv^2}{2\pi r}$ (2) $\frac{mv^2}{r}$
(3) $\frac{2\sqrt{3}mv^2}{\pi r}$ (4) $\frac{3\sqrt{3}mv^2}{4\pi r}$

12. The tension in the string connected between blocks is :-

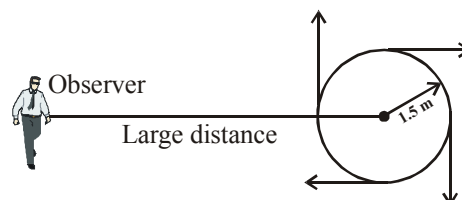


(1) $\frac{80}{3}$ N (2) $\frac{40}{3}$ N
(3) $\frac{50}{7}$ N (4) 26 N

13. A motor car blowing a horn of frequency 124 vib/sec moves with a velocity 20 m/s towards a tall wall. The frequency of the reflected sound heard by the driver will be (velocity of sound in air is 330 m/s)

(1) 109 vib/sec
(2) 132 vib/sec
(3) 140 vib/sec
(4) 248 vib/sec

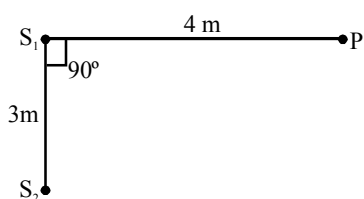
14. A whistle emitting a sound of frequency 440 Hz is tied to a string of 1.5 m length and rotated with an angular velocity of 20 rad/sec in the horizontal plane. Then the range of frequencies heard by an observer stationed at a large distance from the whistle will ($v = 330 \text{ m/s}$) :-



(1) 400.0 Hz to 484.0 Hz
(2) 403.3 Hz to 480.0 Hz
(3) 400.0 Hz to 480.0 Hz
(4) 403.3 Hz to 484.0 Hz

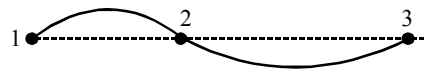
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15. Two tuning forks of frequency 250 Hz and 256 Hz produce beats. If a maximum is observed just now, after how much time the next maximum is observed at the same place ?
 (1) $1/18$ sec (2) $1/24$ sec
 (3) $1/6$ sec (4) $1/12$ sec
16. When a tuning fork A of unknown frequency is sounded with another tuning fork B of frequency 256 Hz, then 3 beats per second are observed. After that A is loaded with wax and sounded with tuning fork B, again 3 beats per second are observed. The frequency of the tuning fork A is :-
 (1) 250 Hz (2) 253 Hz
 (3) 259 Hz (4) 262 Hz
17. Two interfering waves have intensities in the ratio 9:1. Then the ratio of maximum to maximum intensity is :-
 (1) 10 : 8 (2) 4 : 2
 (3) 100 : 64 (4) 16 : 4
18. S_1 and S_2 are two coherent sources of sound of frequency 110 Hz each. They have no initial phase difference. The intensity at a point P due to S_1 is I_0 and due to S_2 is $4I_0$. If the velocity of sound is 330 m/s then the resultant intensity at P is :-



- (1) I_0 (2) $9I_0$ (3) $3I_0$ (4) $8I_0$

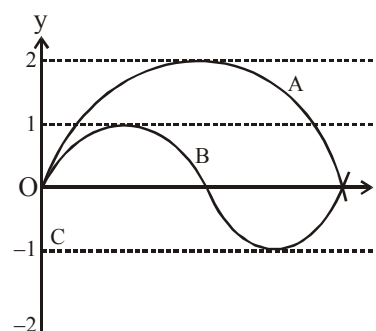
19. The diagram below shows an instantaneous position of a string as a transverse progressive wave travels along it from left to right



Which one of the following correctly shows the direction of the velocity of the points 1, 2 and 3 on the string ?

- | 1 | 2 | 3 |
|-------------------|---------------|---------------|
| (1) $\rightarrow$ | $\rightarrow$ | $\rightarrow$ |
| (2) $\rightarrow$ | $\leftarrow$ | $\rightarrow$ |
| (3) $\downarrow$ | $\downarrow$ | $\downarrow$ |
| (4) $\downarrow$ | $\uparrow$ | $\downarrow$ |

20. The displacement-time graphs for two sound waves A and B are shown in the figure, then the ratio of their intensities I_A/I_B is equal to :-



- (1) 1 : 4 (2) 1 : 16
 (3) 1 : 2 (4) 1 : 1

21. The ratio of the speed of sound in nitrogen gas to that in helium gas, at 300 K is :-

- (1) $\sqrt{2/7}$ (2) $\sqrt{1/7}$
 (3) $\sqrt{3/5}$ (4) $\sqrt{6/5}$

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- 22.** A transverse wave on a string with a linear mass density of 0.20 kg/m is described by the following equation $y = 0.05\sin(420t - 21.0x)$ where x and y are in metres and t is seconds. The tension in the string is equal to ?
 (1) 32 N (2) 42 N
 (3) 66 N (4) 80 N
- 23.** In a stationary wave all the particles :-
 (1) On either side of a node vibrate in same phase
 (2) In the region between two nodes vibrate in same phase
 (3) In the region between two antinodes vibrate in same phase
 (4) Of the medium vibrate in same phase
- 24.** The particle displacement (in cm) in a stationary wave is given by $y(x, t) = 2 \sin(0.1\pi x) \cos(100\pi t)$. The distance between a node and the next antinode is :-
 (1) 2.5 cm (2) 7.5 cm
 (3) 5 cm (4) 10 cm
- 25.** Two wires are kept tight between the same pair of support. The tensions in the wires are in the ratio $2 : 1$, the radii are in the ratio $3 : 1$ and the densities are in the ratio $1 : 2$. The ratio of their fundamental frequencies is :-
 (1) $2 : 3$ (2) $2 : 4$
 (3) $2 : 5$ (4) $2 : 6$
- 26.** An organ pipe P_1 closed at one vibrating in its first overtone and another pipe P_2 open at both ends vibrating in third overtone are in resonance with a given tuning fork. The ratio of the length of P_1 to that of P_2 is :-
- (1) $8/3$ (2) $3/8$
 (3) $1/2$ (4) $1/3$
- 27.** Two coherent sources of different intensities send waves which interfere. The ratio of the maximum intensity to the minimum intensity is 25. The intensities are in the ratio :-
 (1) $25 : 1$ (2) $5 : 1$
 (3) $9 : 4$ (4) $625 : 1$
- 28.** A wave is reflected from a rigid support. The change in phase on reflection will be
 (1) $\pi/4$ (2) $\pi/2$
 (3) π (4) 2π
- 29.** There is a destructive interference between the two waves of wavelength λ coming from two different paths at a point. To get maximum sound or constructive interference at that point, the path of one wave is to be increased by
 (1) $\frac{\lambda}{4}$ (2) $\frac{\lambda}{2}$
 (3) $\frac{3\lambda}{4}$ (4) λ
- 30.** If two waves having amplitudes $2A$ and A and same frequency and velocity, propagate in the same direction in the same phase, the resulting amplitude will be
 (1) $3A$ (2) $\sqrt{5}A$
 (3) $\sqrt{2}A$ (4) A

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31. If the ratio of amplitude of wave is 2 : 1, then the ratio of maximum and minimum intensity is

- (1) 9 : 1 (2) 1 : 9
 (3) 4 : 1 (4) 1 : 4

32. The amplitude of a wave represented by displacement equation

$$y = \frac{1}{\sqrt{a}} \sin \omega t \pm \frac{1}{\sqrt{b}} \cos \omega t \text{ will be}$$

- (1) $\frac{a+b}{ab}$ (2) $\frac{\sqrt{a} + \sqrt{b}}{ab}$
 (3) $\frac{\sqrt{a} \pm \sqrt{b}}{ab}$ (4) $\sqrt{\frac{a+b}{ab}}$

33. A tuning fork of frequency 480 Hz produces 10 beats per second when sounded with a vibrating sonometer string. What must have been the frequency of the string if a slight increase in tension produces lesser beats per second than before

- (1) 460 Hz (2) 470 Hz
 (3) 480 Hz (4) 490 Hz

34. Two tuning forks have frequencies 380 and 384 Hz respectively. When they are sounded together, they produce 4 beats. After hearing the maximum sound, how long will it take to hear the minimum sound

- (1) $\frac{1}{2} \text{ sec}$ (2) $\frac{1}{4} \text{ sec}$
 (3) $\frac{1}{8} \text{ sec}$ (4) $\frac{1}{16} \text{ sec}$

35. A wave travelling along a string is described by $y = 0.005 \sin (80x - 3t)$ in SI units. The wavelength and frequency of the wave are

- (1) $\frac{\lambda}{5} \text{ m}$ and 0.12 Hz
 (2) $\frac{\lambda}{10} \text{ m}$ and 0.24 Hz
 (3) $\frac{\pi}{40} \text{ m}$ and 0.48 Hz
 (4) $\frac{\pi}{20} \text{ m}$ and 0.96 Hz

36. The velocity of sound in air is 360 ms⁻¹ while that in water is 1440 ms⁻¹. If the wavelength of sound in air is 2 cm, that in water is

- (1) $\frac{1}{2} \text{ cm}$ (2) 2 cm
 (3) 8 cm (4) 16 cm

37. A longitudinal wave is represented by

$$x = x_0 \sin 2\pi \left(nt - \frac{x}{\lambda} \right)$$

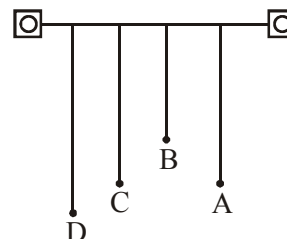
The maximum particle velocity will be four times the wave velocity if

- (1) $\lambda = \frac{\pi x_0}{4}$ (2) $\lambda = 2\pi k_0$
 (3) $\lambda = \frac{\pi x_0}{2}$ (4) $\lambda = 4\pi x_0$

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38. When a wave passes from one medium to another, there is change in
 (1) frequency and velocity
 (2) frequency and wavelength
 (3) wavelength and velocity
 (4) frequency, wavelength and velocity
39. The equation of a transverse wave is given by $y = 10 \sin \pi(0.01x - 2t)$ where y and x are in cm and t in sec. Its frequency is :-
 (1) 10 Hz (2) 2 Hz
 (3) 1 Hz (4) 0.01 Hz
40. Two waves can be expressed as
 $x = a \sin\left(\omega t + \frac{\pi}{6}\right)$ and $x' = a \cos \omega t$. Phase difference between the two waves is :-
 (1) $\frac{\pi}{3}$ (2) $\frac{\pi}{6}$ (3) $\frac{\pi}{2}$ (4) π
41. A wave is represented by the equation;
 $y = A \sin [10\pi x + 15\pi t + (\pi/3)]$
 where x is in m and t in s. The expression represents :-
 (i) A wave travelling in positive x -direction with velocity 1.5 m/s
 (ii) A wave travelling in negative x -direction with velocity 1.5 m/s
 (iii) A wave travelling in negative x -direction with wavelength 0.2 m.
 (iv) A wave travelling in positive x -direction with wavelength 0.2 m
 (1) (i, iii) (2) (ii, iii)
 (3) (iii, iv) (4) (i, iv)

42. The temperature at which speed of sound in air becomes double of its value at 27°C :
 (1) 54°C (2) 327°C
 (3) 927°C (4) -123°C
43. Two tuning forks P and Q when set vibrating, give 4 beats/s. If a prong of the fork P is filled, the beats are reduced to 2/s. What is the frequency of P, if that of Q is 250 Hz ?
 (1) 246 Hz (2) 250 Hz
 (3) 254 Hz (4) 252 Hz
44. Three sound waves of equal amplitudes have frequencies $(v-1)$, v , $(v+1)$. They superpose to give beats. The number of beats produced per second will be :-
 (1) 2 (2) 1
 (3) 4 (4) 3
45. Four pendulums A, B, C and D are suspended from the same elastic support as shown in figure. A and C are of the same length, while B is smaller than A and D is larger than A. If A is given a transverse displacement.



- (1) D will vibrate with maximum amplitude
 (2) C will vibrate with maximum amplitude
 (3) B will vibrate with maximum amplitude
 (4) All the four will oscillate with equal amplitude

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46. Orthoboric acid behaves as a weak monobasic acid giving H_3O^+ and :-
 (1) $[\text{B}(\text{OH})_4]^+$ (2) H_2BO_2^+
 (3) $[\text{B}(\text{OH})_4]^-$ (4) H_2BO_2^-
47. Gallium has smaller atomic radius than aluminium because :-
 (1) poor shielding power of s-electrons of Ga atom
 (2) greater shielding power of s-electrons of Al atom
 (3) greater shielding power of d-electrons of Ga atom
 (4) poor shielding power of d-electrons of Ga atom
48. When an inorganic compounds (X) having $3\text{C}-2\text{e}^-$ as well as $2\text{C}-2\text{e}^-$ bonds reacts with ammonia gas at a certain temperature, gives a compound (Y), isostructural with benzene. Compound (X) with ammonia at a high temperature produces a substance (Z). Which of the following alternative is wrong?
 (1) (X) is diborane(B_2H_6)
 (2) (Y) is $\text{B}_3\text{N}_3\text{H}_6$ (inorganic benzene)
 (3) (Z) is known as inorganic graphite
 (4) (Z) is soft like graphite
49. Consider the following statements for diborane :-
 (i) Boron is sp^3 hybridised
 (ii) $\text{B}-\text{H}-\text{B}$ bond angle is 180°
 (iii) there are two terminal $\text{B}-\text{H}$ bonds for each boron atom
 (iv) there are 12 bonding electrons available
- Of these statements:
 (1) I, III and IV are correct
 (2) I, II and IV are correct
 (3) I, II and III are correct
 (4) II, III and IV are correct
50. In the structure of diborane :-
 (1) all hydrogen atoms lie in one plane and boron atoms lie in a plane perpendicular to this plane
 (2) 2 boron atoms and 4 terminal hydrogen atoms lie in the same plane and two bridging hydrogen atoms lie in the perpendicular plane
 (3) 4 bridging hydrogen atoms and boron atoms lie in one plane and two terminal hydrogen atoms lie in a plane perpendicular to this plane
 (4) all the atoms are present in the same plane
51. Carbon atoms in diamond are bonded with each other in a configuration :-
 (1) linear
 (2) planar
 (3) tetrahedral
 (4) octahedral
52. Which of the following is correct?
 (1) silicones are organosilico polymers containing $\text{Si}-\text{O}-\text{Si}$ linkage
 (2) R_3SiCl on hydrolysis gives $\text{R}_3\text{Si}-\text{O}-\text{SiR}_3$
 (3) Both of these
 (4) None of these

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53. In silicon dioxide :-

- (1) each silicon atom is surrounded by four oxygen atoms and each oxygen atom is bonded to two silicon atoms
- (2) each silicon atom is surrounded by two oxygen atoms and each oxygen atom is bonded to two silicon atoms
- (3) silicon atom is bonded to two oxygen atoms
- (4) there are double bonds between silicon and oxygen atoms

54. Ge (II) compounds are powerful reducing agents where Pb(IV) compounds are strong oxidants. It can be due to:

- (1) Pb is more electronegative than Ge
- (2) Ionisation energy of lead is less than that of Ge
- (3) Ionic radii of Pb^{2+} and Pb^{4+} are larger than those of Ge^{2+} and Ge^{4+}
- (4) more pronounced inert pair effect in lead than in Ge.

55. Buckminster fullerene is :-

- (1) graphite
- (2) C_{60}
- (3) diamond
- (4) bone charcoal

56. $2RCl + Si \xrightarrow[570\text{ K}]{\text{Cu powder}} R_2SiCl_2 \xrightarrow{H_2O} R_2Si(OH)_2$

$\xrightarrow{\text{Polymerisation}} (A)$. (A) is :-

- (1) cyclic silicone
- (2) cross linked silicone
- (3) linear silicone
- (4) none of these

57. The hybridisation states of boron and nitrogen in borazole are respectively :-

- (1) sp, sp
- (2) sp, sp^2
- (3) sp^2 , sp^2
- (4) sp^2 , sp^3

58. Thallium shows +1 oxidation state because:

- (1) of its high reactivity
- (2) of inert pair effect
- (3) of its amphoteric nature
- (4) it is similar to silver

59. C—C bond length is maximum in:

- (1) diamond
- (2) graphite
- (3) fullerene
- (4) naphthalene

60. The pairs of species of oxygen and their magnetic behaviour are noted below. Which of the following represents the correct description?

- (1) O_2^- , O_2^{2-} —Both diamagnetic
- (2) O^+ , O_2^{2-} —Both paramagnetic
- (3) O_2^+ , O_2 —Both paramagnetic
- (4) O , O_2^{2-} —Both paramagnetic

61. Which of the following represents the Lewis structure of N_2 molecule?

- (1) $\overset{\times}{\underset{\times}{N}} \equiv \overset{\times}{\underset{\times}{N}}$
- (2) $\overset{\times}{\underset{\times}{N}} \equiv \overset{\times}{\underset{\times}{N}}$
- (3) $\overset{\times}{\underset{\times}{N}} - \overset{\times}{\underset{\times}{N}}$
- (4) $\overset{\times}{\underset{\times}{N}} = \overset{\times}{\underset{\times}{N}}$

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62. Hydrogen bond is strongest in:

- (1) S—H—O
- (2) O—H—S
- (3) F—H—F
- (4) O—H—N

63. Which one of the following arrangements of molecules is correct on the basis of their dipole moments?

- (1) $\text{BF}_3 > \text{NH}_3 > \text{NF}_3$
- (2) $\text{BF}_3 > \text{NF}_3 > \text{NH}_3$
- (3) $\text{NH}_3 > \text{BF}_3 > \text{NF}_3$
- (4) $\text{NH}_3 > \text{NF}_3 > \text{BF}_3$

64. Match List-I with List-II

List-I	List-II
A. CO_2	1. Pyramid
B. SO_2	2. Tetrahedral
C. BF_3	3. Triangular planar
D. CH_4	4. Linear
	5. Angular

The correct match is :

A	B	C	D
(1) 4	5	3	2
(2) 5	4	3	2
(3) 4	3	2	1
(4) 1	2	3	4

65. Which of the following statements is not correct from the view point of molecular orbital theory?

- (1) Be_2 is not a stable molecule
- (2) He_2 is not stable but He_2^+ is expected to exist
- (3) Bond strength of N_2 is maximum amongst the homonuclear diatomic molecules belonging to second period
- (4) The order of energies of molecular orbitals in N_2 molecule is $\rightarrow$

$$\sigma 2s < \sigma^* 2s < \sigma 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$$

66. Which of the following is not a correct statement?

- (1) Every AB_5 molecule does in fact have square pyramid structure.
- (2) Multiple bonds are always shorter than corresponding single bonds
- (3) The electrons deficient compounds can act as Lewis acids
- (4) H_2O molecule has 'V' shape

67. Which of the following species has a linear shape?

- (1) NO_2^+
- (2) O_3
- (3) NO_2^-
- (4) SO_2

68. Which of the following has $p\pi-d\pi$ bonding?

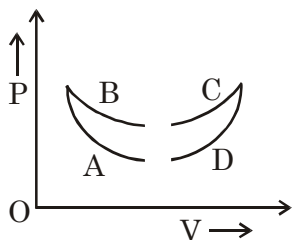
- (1) NO_3^-
- (2) SO_3^{2-}
- (3) BO_3^{3-}
- (4) CO_3^{2-}

69. In OF_2 , number of bond pairs and lone pairs of electrons are respectively :

- (1) 2, 6
- (2) 2, 8
- (3) 2, 10
- (4) 2, 9

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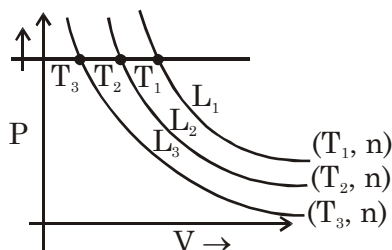
70. Four curves A, B, C and D are drawn in figure for a given amount of gas. The curve which represents adiabatic and isothermal changes, respectively, is :-



- (1) C and D (2) D and C
(3) A and B (4) B and A
71. Which of the following methods for the calculation of heat of a reaction is not correct?

- (1) $\Delta H_{\text{reaction}}^{\ominus} = \sum \Delta_f H_{\text{products}}^{\ominus} - \sum \Delta_f H_{\text{reactants}}^{\ominus}$
(2) $\Delta H_{\text{reaction}}^{\ominus} = \sum (\text{BE})_{\text{reactants}} - \sum (\text{BE})_{\text{products}}$
(3) $\Delta H_{\text{reaction}}^{\ominus} = \sum \Delta_{\text{comb}} H_{\text{(reactants)}}^{\ominus} - \sum \Delta_{\text{comb}} H_{\text{(products)}}^{\ominus}$
(4) $\Delta H_{\text{reaction}}^{\ominus} = \sum \Delta_{\text{solution}} H_{\text{(reactants)}}^{\ominus} - \sum \Delta_{\text{solution}} H_{\text{(products)}}^{\ominus}$

72. Boyle's Law for an ideal gas can be plotted as shown ($\rightarrow$) (n : moles T : temperature)
Note : T and n are kept constant along line L_1 , L_2 , and L_3 .



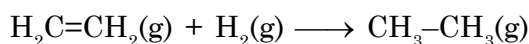
If follows from the above graph :-

- (1) $T_1 > T_2 > T_3$ (2) $T_1 < T_2 < T_3$
(3) $T_1 = T_2 = T_3$ (4) None of these

73. According to the kinetic theory of gases, for a diatomic molecule :-

- (1) The pressure exerted by the gas is proportional to the mean velocity of the molecule.
(2) The pressure exerted by the gas is proportional to the root mean velocity of the molecule.
(3) The root mean square velocity of the molecule is inversely proportional to the temperature.
(4) The mean translational kinetic energy of the molecule is proportional to the absolute temperature

74. Calculate the enthalpy of the following reaction :



The bond energies of C-H, C-C, C=C, and H-H are 99, 83, 147, and 104 kcal, respectively :-

- (1) 30 kcal (2) -30 kcal
(3) 60 kcal (4) -60 kcal

75. Which one of the following is a state property/function?

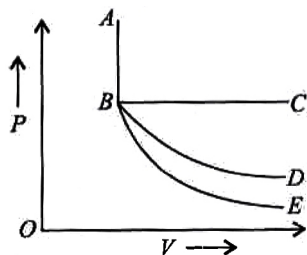
- (1) Heat
(2) Work
(3) Internal energy
(4) Heat capacity

76. Which of the following is an extensive property?

- (1) Enthalpy
(2) Concentration
(3) Density
(4) Viscosity

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77. In P-V diagram shown below,

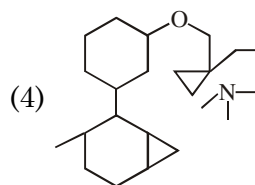
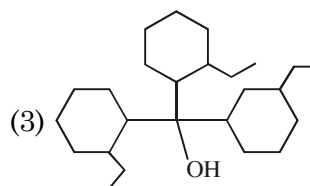
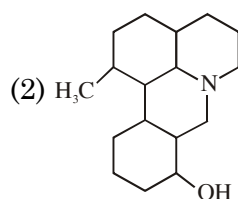
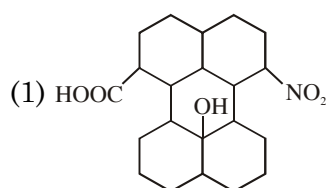


- (1) AB represents adiabatic process.
- (2) AB represents isothermal process.
- (3) AB represents isobaric process.
- (4) AB represents isochoric process.

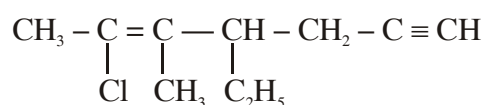
78. Which of the following acid will release maximum amount of heat when completely neutralised by strong base NaOH?

- (1) 1 M HCl
- (2) 1 M HNO₃
- (3) 1 M HClO₄
- (4) 1 M H₂SO₄

79. Which of the following is a Heterocyclic compound.

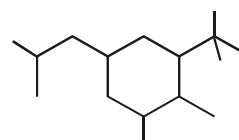


80. The IUPAC name of



- (1) 2-Chloro-4-ethyl-3-methyl-6-heptyn-2-ene
- (2) 2-Chloro-4-ethyl-3-methyl-2-heptyn-6-yne
- (3) 6-Chloro-4-ethyl-5-methyl-1-heptyn-5-ene
- (4) 6-Chloro-4-ethyl-5-methyl-5-hepten-1-yne

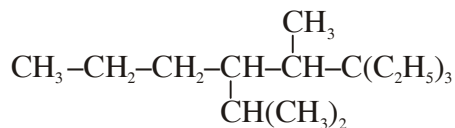
81. Number of 1°, 2°, 3° and 4° C present in compound are respectively



- (1) 7, 4, 5 and 1
- (2) 7, 3, 4 and 1
- (3) 7, 3, 5 and 1
- (4) 7, 4, 4 and 1

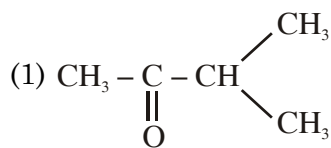
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82. The correct IUPAC name of the compound is :

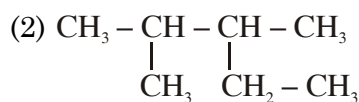


- (1) 3,3-Diethyl-4-methyl 5-(1-methyl ethyl) octane
- (2) 6,6-Diethyl-4-methyl-5-isopropyloctane
- (3) 6,6-Diethyl-3-methyl 5-(1-methylethyl) octane
- (4) 6,6-Diethyl-4-isopropyl-5-methyloctane

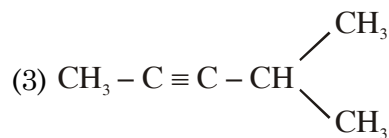
83. The incorrect IUPAC name is:



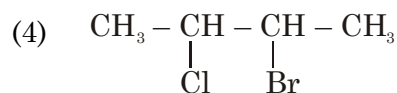
2-Methyl-3-butanone



2,3-Dimethylpentane

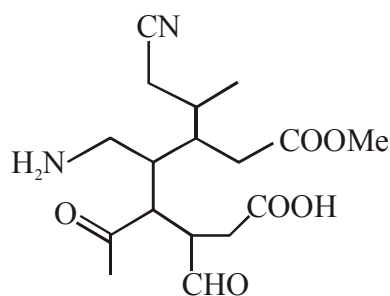


4-Methyl-2-pentyne



2-bromo-3-chloro butane

84. Number of functional groups and double bond equivalent are respectively.



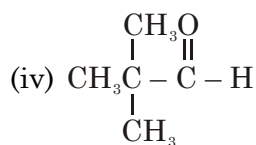
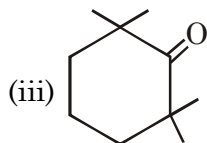
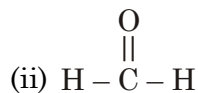
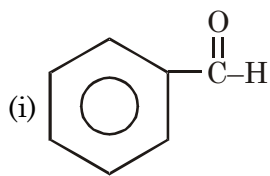
- (1) 6, 1 (2) 6, 5
- (3) 6, 6 (4) 6, 7

85. Which of the following is pair of homologues?

- (1) CH_3OH , $\text{CH}_3-\text{O}-\text{CH}_3$
- (2) CH_3OH , CH_3SH
- (3) $\text{CH}_3-\text{O}-\text{CH}_3$, $\text{CH}_3-\text{CH}_2-\text{OH}$
- (4) CH_3OH , $\text{CH}_3\text{CH}_2\text{OH}$

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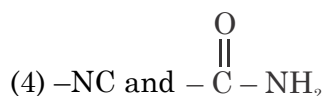
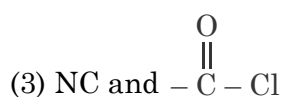
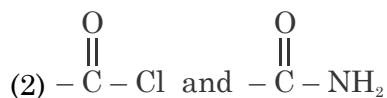
86. Which of the following compounds does not have an alpha hydrogen?



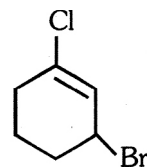
- (1) i
 (2) ii
 (3) iii
 (4) All of these

87. Isocyano and carbamoyl prefix is used respectively for :-

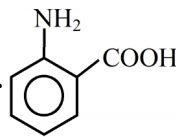
- (1) $-\text{CHO}$ and $-\text{COR}$



88. The IUPAC name of the compound is



- (1) 6-bromo-2-chlorocyclohexene
 (2) 3-bromo-1-chlorocyclohexene
 (3) 1-bromo-3-chlorocyclohexene
 (4) 2-bromo-6-chlorocyclohex-1-ene

89. The IUPAC name of  is -

- (1) 1-amino-2-carboxybenzene
 (2) 2-amino-1-carboxybenzene
 (3) 1-amino-2-benzenecarboxylic acid
 (4) 2-aminobenzenecarboxylic acid

90. IUPAC name of $\text{CH}_3 - \text{CH} = \text{CH} - \text{C} \equiv \text{CH}$ is :-

- (1) 2-Penten-4-yne
 (2) 1-Pentyn-3-ene
 (3) 3-Penten-1-yne
 (4) None of these

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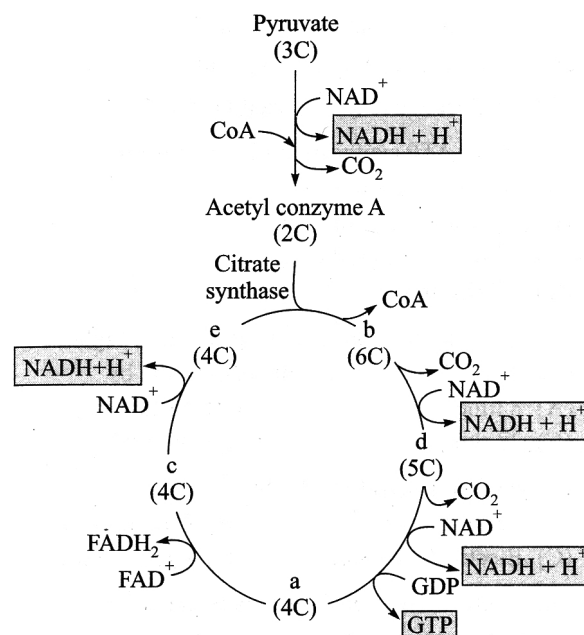
91. Which of the following statement is incorrect ?
- Glycolysis occurs in cytoplasm of the cell
 - Generally availability of CO_2 is not a limiting factor for C_4 plants
 - ETS occurs in outer mitochondrial membrane
 - Aerobic respiration occurs inside mitochondria
92. Which of the following is representing the correct set of inputs in dark reaction- C_3 pathway, for fixation of 1 molecules of CO_2 ?
- ATP-3; NADPH-2
 - ATP-12; NADPH-12
 - ATP-18; NADPH-12
 - ATP-30; NADPH-12
93. Which component of ETC is not a protein?
- Cytochrome
 - Ubiquinone
 - Cytochrome oxidase
 - All the above
94. $\text{NADP}^+ + \text{e}^- + \text{H}^+ \xrightarrow{\text{A}} \text{NADPH}^+$, Enzyme A is :-
- NADP oxidase
 - NADP reductase
 - Transacetylase
 - ATPase
95. Find out the values of x, y and z in the given equation :
- $$2(\text{C}_{51}\text{H}_{98}\text{O}_6) + x\text{O}_2 \rightarrow y\text{CO}_2 + z\text{H}_2\text{O} + \text{Energy}$$
- x-102, y-145, z-98
 - x-145, y-102, z-98
 - x-98, y-102, z-145
 - x-145, y-98, z-102

96. Read the following statements (A-H) :-
- In PS-I, Chlorophyll-b has an absorption peak at 700 nm.
 - Reaction centre is the part of light harvesting complex.
 - Movement of electrons from acceptor to PS I is uphill transport.
 - In light harvesting complex, pigment molecules are bound to proteins.
 - Reaction centre is formed by many molecules of chlorophyll-a only.
 - Photorespiration might be photoprotective in function
 - Rate of transpiration is maximum in blue light.
 - Rate of photosynthesis is maximum in red light.

How many of the above statements are correct?

- 3
- 4
- 5
- 6

97. Recognise the figure and find out the correct matching.



- b-malic acid, d-OAA, e- α -ketoglutaric acid, c-succinic acid, a-citric acid
- a-malic acid, c-OAA, b- α -ketoglutaric acid, d-succinic acid, e-citric acid
- c-malic acid, e-OAA, d- α -ketoglutaric acid, a-succinic acid, b-citric acid
- e-malic acid, a-OAA, c- α -ketoglutaric acid, b-succinic acid, d-citric acid

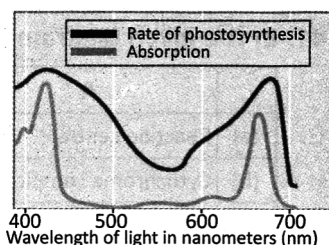
98. In PS – I, if the number of pigment molecules is n , then what will be the number of pigment molecules in its antennae?

- (1) $n/2$ (2) $n - 1$
(3) $n + 1$ (4) $2n - 1$

99. Fill in the blanks :

- ATP synthase consists of two major components, CF_0 and CF_1 . Thea.... headpiece is peripheral membrane protein complex and contains the site for synthesis of ATP from ADP and inorganic phosphate.
 -b.... is an integral membrane protein complex that forms the channel through which proton crosses the membrane.
 - For each ATP produced,c.... passes throughb.... from the ...d.... to thee.... down the electrochemical potential gradient.
- (1) a- CF_0 , b- CF_1 , c- $2H^+$, d-matrix, e-inner membrane space
(2) a- CF_1 , b- CF_0 , c- $2H^+$, d-stroma, e-lumen
(3) a- CF_0 , b- CF_1 , c- $3H^+$, d-intermembrane space, e-matrix
(4) a- CF_1 , b- CF_0 , c- $3H^+$, d-lumen, e-stroma

100. Adjacent graph is indicating rate of absorption of light and rate of photosynthesis in visible region. Which of the following is correct extract?



- (1) Rate of absorption of light and rate of photosynthesis have no inter-relation
(2) Rate of absorption of light and rate of photosynthesis nearly overlap

- (3) Rate of absorption of light and rate of photosynthesis point to point overlap
(4) Rate of absorption of light and rate of photosynthesis have inverse relationship

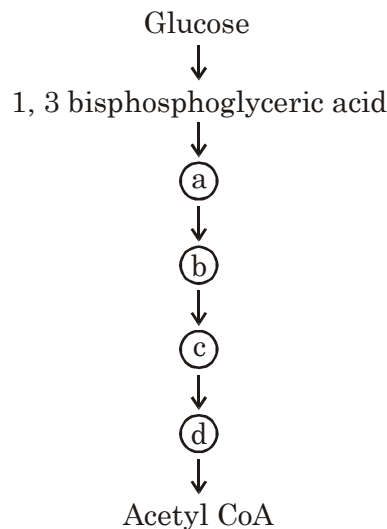
101. Optimum CO_2 concentration for sugarcane and wheat are :-

- (1) $360 \mu\ell L^{-1}$ and $450 \mu\ell L^{-1}$
(2) $450 \mu\ell L^{-1}$ and $360 \mu\ell L^{-1}$
(3) $360 \mu\ell L^{-1}$ and $360 \mu\ell L^{-1}$
(4) $450 \mu\ell L^{-1}$ and $450 \mu\ell L^{-1}$

102. Most crucial step in calvin-benson-bassham cycle is :-

- (1) Regeneration
(2) Reduction
(3) Carboxylation
(4) Both (1) and (3)

103. Reocognise the figure and find out the correct matching :-



- (1) b-2-phosphoglycerate, a-3-phosphoglyceric acid, c-pyruvic acid, d-phosphoenolpyruvate
(2) a-2-phosphoglycerate, b-3-phosphoglyceric acid, d-pyruvic acid, c-phosphoenolpyruvate
(3) a-3-phosphoglycerate, b-2-phosphoglyceric acid, d-pyruvic acid, c-phosphoenolpyruvate
(4) a-3-phosphoglycerate, b-2-phosphoglyceric acid, d-pyruvic acid, c-Acetoacetate

104. Read the following statements and find out the incorrect statement :-

- Ultimately all the food that is respired for life processes comes from photosynthesis.
- The energy released by oxidation in respiration is directly used for the life processes.
- The carbon skeleton produced during respiration is used as precursors for biosynthesis of other molecules in cell.
- Plants, unlike animals, have no specialised organs for gaseous exchange but they have stomata and lenticels for this purpose.
- During oxidation within a cell, all the energy contained in respiratory substrates is released free into the cell in a single step, which is trapped as chemical energy in the form of ATP.

- (1) b and e (2) a and b
- (3) c and e (4) b and d

105. Site of TCA cycle in aerobic bacteria :-

- (1) Cytoplasm
- (2) Cell membrane
- (3) Mesosomes
- (4) Both (2) and (3)

106. Match the following columns :

Column I		Column II	
A.	4-C- compound	1.	Acetyl CO-A
B.	2-C- compound	2.	Pyruvate
C.	5-C- compound	3.	Citric acid
D.	3-C- compound	4.	α -ketoglutaric acid
		5.	Malic acid

Codes :

- | | | | |
|-------|---|---|---|
| A | B | C | D |
| (1) 2 | 5 | 3 | 1 |
| (2) 3 | 1 | 4 | 2 |
| (3) 5 | 1 | 4 | 2 |
| (4) 5 | 3 | 1 | 2 |

107. α -ketoglutarate of Kreb's cycle produces an important amino acid called

Fill up the blank with the given option.

- (1) succinate
- (2) glycine
- (3) glutamate
- (4) alanine

108. Correct sequence of electron acceptor of ATP synthesis is :-

- (1) cty - a, a_3 , b, c
- (2) cyt-b, c, a, a_3
- (3) cyt-b, c, a_3 , a
- (4) cyt-c, b, a, a_3

109. Yeast poison themselves to death when the concentration of alcohol reaches :-

- (1) 20% (2) 13%
- (3) 15% (4) 14%

110. Statement A : In certain biochemical reaction two redox equivalents are removed in the form of two hydrogen atom from PGAL and transferred to a molecule of NAD^+ .

Statement B : The site of this reaction is mitochondrial matrix.

- (1) Only A is correct
- (2) Only B is correct
- (3) Both A and B are correct
- (4) Both A and B are incorrect

111. Statement A : Less than 7% of energy in glucose is released and not all of it trapped as high energy bonds of ATP in alcoholic fermentation.

Statement B : $NADH$ is oxydized to NAD^+ in lactic acid fermentation only.

- (1) A is correct
- (2) B is correct
- (3) Both are correct
- (4) Both are incorrect

112. Which of the following are mobile electron carriers present in the inner membrane of mitochondria?

- (1) Ubiquinone and Cyt a_3
- (2) Cyt b and Cyt c_1
- (3) Cyt c and Cyt a
- (4) Ubiquinone and Cyt c

113. Oxidation product of photosynthesis in purple and green sulphur bacteria is :-

- (1) O_2
- (2) H_2O
- (3) Sulphate/sulphur
- (4) Carbohydrates

114. What are the reasons why plants can get along without respiratory organs?

- a. Each plant part takes care of its own gas exchange needs. There is very little transport of gases from one plant part to another.
- b. Plants do not present great demands for gaseous exchanges; root, stem and leaves respire at rates far lower than animals do.
- c. The distance that gases must diffuse even in large, bulky plants is not great.

- (1) a and b
- (2) b and c
- (3) c and a
- (4) a, b and c

115. In half leaf experiment,A..... was used to absorb / remove CO_2 around the ...B... half leaf while another half showed theC... of starch.

	A	B	C
(1)	$K_2Cr_2O_7$	Exposed	Presence
(2)	$K_2Cr_2O_7$	Enclosed	Absence
(3)	KOH	Exposed	Absence
(4)	KOH	Enclosed	Presence

116. The rate of respiration increases for every rise of $10^\circ C$ temperature :-

- (1) 2 times
- (2) 2–3 times
- (3) can not predict rate of respiration
- (4) rate of respiration is not affected by temperature

117. Match the Column I with Column II :-

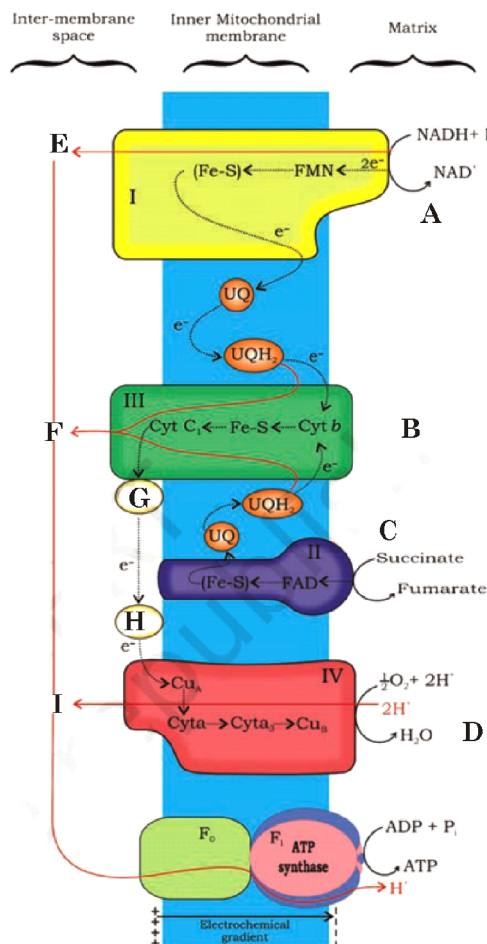
Column I (Scientists/Physiologists)		Column II (Contribution)	
1.	Priestley	A.	Light is essential for photosynthesis
2.	Englemann	B.	Evolved O_2 comes from H^+ donor
3.	Ingenhousz	C.	Action spectrum of photosynthesis
4.	Van Niel	D.	Discovery of O_2

- (1) 1-B, 2-D, 3-C, 4-A
- (2) 1-B, 2-C, 3-D, 4-A
- (3) 1-D, 2-C, 3-A, 4-B
- (4) 1-D, 2-A, 3-C, 4-B

118. Select incorrect statement :-

- (1) Light saturation occurs at 10% of full sunlight
- (2) NADPH is not synthesized in photorespiration
- (3) Temperature optimum is lower for C_4 plants than C_3 plants
- (4) CO_2 is usually a major limiting factor in photosynthesis

119. Chemiosmotic synthesis of ATP is shown in the diagram in mitochondrion. Identify correct match :-



- (1) A – NADH dehydrogenase,
B – Succinate dehydrogenase,
G – Cyt C,
H – Cyt b,
E – 4H^+
- (2) D – Cyt C oxidase,
B – Cyt C reductase,
A – Complex I,
H – Mobile carrier,
I – 2H^+
- (3) E – 4H^+ ,
F – 4H^+ ,
I – 2H^+ ,
C – Succinate dehydrogenase,
B – NADP reductase
- (4) H – Cyt C,
A – NADH oxidase,
B – Cyt C oxidase,
D – OEC

120. If maize–A, Bell pepper–B, Tomato–C and sorghum–D, then which option correctly represents the pair of C_3 plants?

- (1) A and B
- (2) B and C
- (3) B and D
- (4) C and D

121. Which of the following option is wrong w.r.t. photorespiration

- (1) It is closely associated with CO_2 compensation point
- (2) It recovers 75% of carbon which would otherwise be lost as glycolate
- (3) It occurs in plant which have high CO_2 compensation point
- (4) It never occur in C_4 plants.

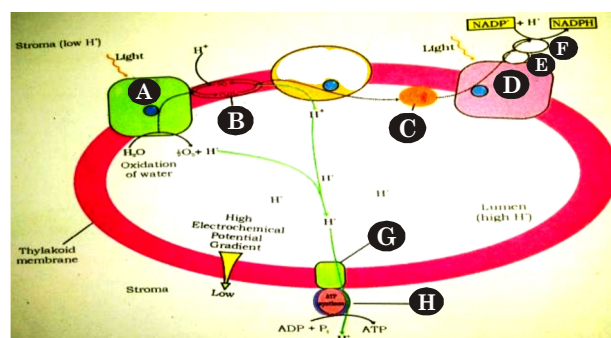
122. RuBisCO is found in :-

- (1) C_3 plants but not in C_4 plants
- (2) C_4 plants but not in C_3 plants
- (3) Both C_3 and C_4 plants
- (4) Neither C_3 plants nor C_4 plants

123. During Calvin cycle, how many ATP are required for regeneration of RuBP?

- (1) 1 ATP
- (2) 2 ATP
- (3) 3 ATP
- (4) 4 ATP

124. Chemiosmotic synthesis of ATP is shown in the diagram in thylakoid. Identify correct match :-



- (1) A – PS I, B – UQH_2 , D – PS II, G – CF_0 ,
F – FNR
- (2) A – PS II, B – PQ, E – Fd, H – CF_1
- (3) F – FNR, C – PC, B – Plastoquinol,
G – ATPase activity
- (4) A – PS II, B – Cyt complex, C – UQ,
D – PS I, E – FRS

125. A sudden change from anaerobic to aerobic process produces :-

- (1) Pasteur effect
- (2) Emerson effect
- (3) Blackman's law
- (4) Warburg effect

126. Which of the following statement is wrong :-

- (1) Acetyl coA is common respiratory intermediate in metabolic processes
- (2) Protein may enter at several steps in respiratory pathway for breakdown
- (3) Krebs's cycle is an amphibolic pathway
- (4) Hexokinase is also called as pacemaker of glycolysis because it is an example of allosteric enzyme.

127. Find the values of w, y and z in the given equation.



↓ mitochondrial matrix



- (1) w-4, y-3, z-1
- (2) w-3, y-2, z-1
- (3) w-3, y-3, z-2
- (4) w-4, y-3, z-2

128. How many ATPs are formed per glucose molecule in Krebs's cycle?

- | | |
|-------|--------|
| (1) 2 | (2) 24 |
| (3) 6 | (4) 28 |

129. Ubiquinone is located

- (1) On the outer surface of inner membrane
- (2) On the inner surface of outer membrane
- (3) Within inner membrane
- (4) Within outer membrane

130. The crucial events in aerobic respiration are :-

1. The complete oxidation of pyruvate by the stepwise removal of all the hydrogen atoms, leaving three molecules of CO_2
2. The passing on of the electrons removal as part of the hydrogen atoms to molecular O_2 with simultaneous synthesis of ATP

(1) First process is ETS and takes place in matrix of the mitochondria while second process is TCA and takes place in on the inner membrane of mitochondria.

(2) First process is TCA and takes place in the matrix of mitochondria while the second process is ETS and is located on the inner membrane of mitochondria

(3) First process is ETS and takes place on the inner membrane of mitochondria while second process is TCA and takes place in the matrix of the mitochondria

(4) First process is TCA and takes place on the inner membrane of mitochondria while second process is ETS and takes place in the matrix of mitochondria

131. Cyanides can inhibit respiration by :-

- (1) blocking complex I
- (2) Destroying complex II
- (3) Preventing electron flow from complex III
- (4) Preventing electron flow from complex IV

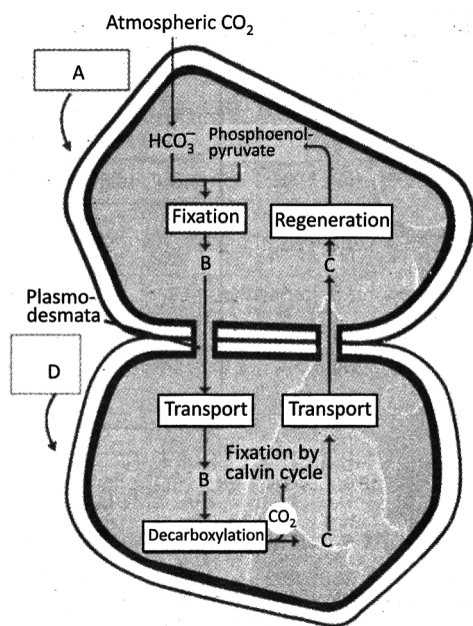
132. Match the columns :-

Column I		Column II	
1.	C ₃ plants	A.	Kranz anatomy
2.	C ₄ plants	B.	Significant photorespiration
		C.	Low optimum temperature
		D.	RuBisCO in mesophyll cells
		E.	Calvin Cycle in bundle sheath cells
		F.	Low CO ₂ fixation rate
		G.	PEP case present in mesophyll cells

- (1) 1-B, C, D, F 2-A, E, G
 (2) 1-B, D, F 2-A, C, E, G
 (3) 1-C, D, E 2-A, B, F, G
 (4) 1-A, D, F 2-B, C, E, G

133. C₄ pathway is shown in the diagram.

Correctly identify A-D.

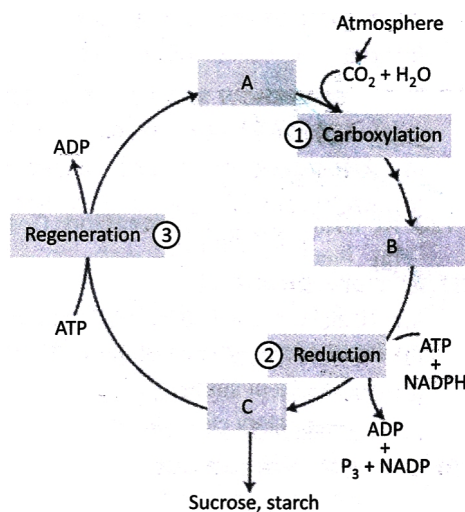


	A	B	C	D
(1)	Bundle sheath cell	C ₄ acid	C ₃ acid	Mesophyll cell
(2)	Epidermis	C ₃ acid	C ₄ acid	Mesophyll cell
(3)	Mesophyll cell	C ₄ acid	C ₃ acid	Bundle sheath cell
(4)	Epidermis	C ₃ acid	C ₄ acid	Bundle sheath cell

134. Red drop in photosynthesis in green algae refer to decrease in the rate of photosynthesis in :-

- (1) Blue light
 (2) Green light
 (3) Light wavelength greater than 680 nm
 (4) Light wavelength less than 680 nm

135. A, B and C are three compounds / intermediates of Calvin cycle as shown. Select the option with correct identification.



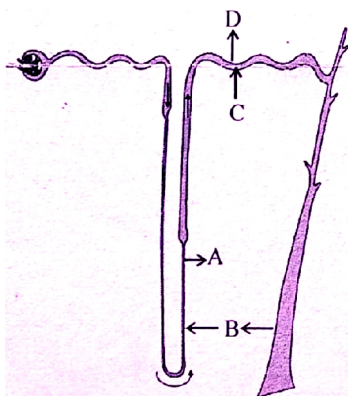
	A	B	C
(1)	Ribulose-1, 5-biphosphate	3-phospho-glycerate	Triose phosphate
(2)	Ribulose-1 5-biphosphate	Triose phosphate	3-phospho-glycerate
(3)	Triose phosphate	Ribulose-1, 6-biophosphate	3-phospho-glycerate
(4)	3-Phospho-glycerate	Ribulose-1 6-biphosphate	Triose phosphate

136. Identify the correct match from the columns I, II and III?

Column-I	Column-II	Column-III
(1) PCT	(a) In medulla of kidney	(i) concentration of urine
(2) Loop of Henle	(b) Simple squamous epithelium	(ii) Main area of absorption
(3) DCT	(c) Brush border	(iii) Ultrafiltration
(4) Bowman's capsule	(d) Aquaporin	(iv) Conditional reabsorption of H_2O , salts

- (1) 2-(a)-(i), 1-(c)-(iv), 3-(d)-(ii), 4-(b)-(iii)
 (2) 1-(c)-(ii), 2-(b)-(i), 3-(d)-(iv), 4-(a)-(iii)
 (3) 2-(a)-(i), 1-(c)-(ii), 4-(b)-(iii), 3-(d)-(iv)
 (4) 3-(d)-(iv), 1-(c)-(i), 2-(a)-(ii), 4-(b)-(iii)

137. Reabsorption and secretion of major substances at different parts of the Nephron are shown in the given figure. In this figure A, B, C, D are?



- (1) HCO_3^- , Nutrients, Urea, NaCl
 (2) NaCl, Urea, K^+ , HCO_3^-
 (3) NaCl, Urea, HCO_3^- , Ammonia
 (4) H_2O Urea, NaCl, HCO_3^-

138. Match the following

Column-I	Column-II
A PCT	i Selective secretion of H^+ and K^+ ions
B DCT	ii 70-80 percent of electrolytes and water reabsorbed
C Descending limb	iii Allow passage of small amount of urea
D Ascending limb	iv permeable to water
E Collecting duct	v permeable to salts

- (1) A-i ; B-ii ; C-iii ; D-iv ; E-v
 (2) A-i ; B-iii; C-iv ; D-ii ; E-v
 (3) A-ii; B-i; C-iv; D-v; E-iii
 (4) A-ii : B-i ; C-v ; D-iv ; E-iii

139. The release of ADH is triggered when the osmoreceptors in the hypothalamus detect on :-

- (1) Decrease in osmolarity of the blood below the set point of 300 mosmL^{-1}
 (2) Increase in osmolarity above a set point of 300 mosmL^{-1}
 (3) Increase in blood volume
 (4) constant osmolarity

140. Consider the following statements each with one or two blanks.

- (i) Towards the centre of the inner concave surface of the kidney is a notch called(a)..... through which ureter, blood vessels and nerves enter.
 (ii) The medulla of kidney is divided into a few conical masses called(b)..... projecting into the(c).....
 (iii) Glomerulus is a tuft of capillaries formed by the(d)..... Blood from the glomerulus is carried away by an(e).... arteriole.

Which one of the following options correctly fills the blanks in any two of the statements ?

- (1) (a) - renal pelvis, (b) - calyces, (c) medullary pyramids
 (2) (b) - medullary pyramids, (c) - calyces, (d) - afferent, (e) - efferent
 (3) (b) - columns of Bertini, (c) - chordae tendinde, (d) efferent, (e) - afferent
 (4) (a) - hilum, (d) - efferent, (e) - afferent

141. Consider the following four statements (A-D) and select the option which includes all correct :-

- (A) Henle's loop plays a significant role in the maintenance of high osmolarity of medullary interstitial fluid.
- (B) Collecting duct plays a role in the maintenance of pH and ionic balance of blood by the selective secretion of H^+ and K^+ ions
- (C) Conditional reabsorption of Na^+ and water takes place in PCT.
- (D) Nearly all of the essential nutrients and 70-80 percent of electrolytes and water are reabsorbed by PCT.

- (1) statements (A), (B), (D)
- (2) Statements (A), (C), (D)
- (3) Statements (B), (C)
- (4) Statements (C), (D)

142. Which one of the following statements is correct with respect to kidney function regulation?

- (1) When someone drinks lot of water, ADH releasees increased.
- (2) Exposure to cold temperature stimulates ADH release.
- (3) A decrease in glomerular blood flow stimulates formation of Angiotensin II.
- (4) During summer when body loses lot of water by evaporation, the release of ADH is decreased.

143. Which one of the following statements are correct :-

- (1) The medullary zone of kidney is divided into a few conical masses called medullary pyramids projecting into the calyces
- (2) Inside the kidney the cortical region extends in between the medullary pyramids
- (3) Glomerulus alongwith Bowman's capsule is called the renal corpuscle
- (4) All the above

144. (i) On an average 25-30 gm of urea is excreted out per day.

(ii) Primary function of sweat is to facilitate a colling effect on the body surface.

(iii) Glycosuria and ketonuria in urine are indicative of diabetes insipidus

(iv) Small amount of nitrogenous wastes could be eliminated through saliva.

How many statements are correct ?

- (1) Four (2) Three (3) Two (4) One

145. Regulation of kidney function of Juxtaglomerular Apparatus (JGA) involves certain steps given below. Arrange them in the correct order :-

(a) Release of enzyme renin

(b) Release of aldosterone from adrenal gland

(c) Reabsorption of Na^+ and water at DCT

(d) Decrease in blood pressure and blood volume

(e) Conversion of angiotensinogen to angiotensin II

(1) a, e, b, c, d

(2) b, c, a, e, d

(3) d, a, b, e, c

(4) d, a, e, b, c

146. Which one of the following statements in regard to the excretion by the human kidney is correct?

(1) Ascending limb of Loop of Henle is impermeable to electrolytes

(2) Descending limb of Loop of Henle is impermeable to water

(3) Distal convoluted tubule is incapable of reabsorbing HCO_3^-

(4) Nearly 99 per cent of the glomerular filtrate is reabsorbed by the renal tubules

- 147.** Amino acids that are participating in Krebs-Henseleit cycle, but are not otherwise found in proteins are
- (1) Ornithine and arginine
 - (2) Ornithine and citrulline
 - (3) Citrulline and arginine
 - (4) Ornithine and aspartic acid
- 148.** Which of the following statements is incorrect?
- (1) Tadpole larva is ammonotelic but adult frog is ureotelic
 - (2) Sharks and rays are uricotelic
 - (3) Ammonia is the most toxic and most soluble waste
 - (4) Animals that lay shelled eggs are uricotelic
- 149.** In Krebs-Henseleit cycle, the step at which urea is synthesized is catalysed by
- (1) Argininosuccinate synthase
 - (2) Argininosuccinate lyase
 - (3) Arginase
 - (4) Ornithine transcarbamoylase
- 150.** Urea is synthesized from
- (1) CO_2 and uric acid
 - (2) CO_2 and ammonia
 - (3) Ammonia and uric acid
 - (4) CO_2 and water
- 151.** Marine teleost fishes excrete a large proportion of their nitrogen as
- (1) Creatine
 - (2) Hippuric acid
 - (3) Trimethylamine oxide (TMO)
 - (4) Ammonia
- 152.** Internal section of Kidney shows extension of cortex into medullary regions between the pyramids and is called
- (1) Ducts of Bellini
 - (2) Major calyx
 - (3) Minor calyx
 - (4) Columns of Bertini
- 153.** The protective layer of Kidney is/are:
- (1) Renal capsule
 - (2) Adipose tissue
 - (3) Renal fascia
 - (4) All of these
- 154.** Which of the following epithelium is present in the inner lining of urinary bladder?
- (1) Columnar
 - (2) Cuboidal
 - (3) Transitional
 - (4) Pseudostratified
- 155.** The structural and functional unit of Kidney is called
- (1) Nephron
 - (2) Uriniferous tubule
 - (3) Both (1) and (2)
 - (4) Malpighian tubule
- 156.** Kidneys are vital organs of the body because they help in
- (1) Removal of metabolic wastes
 - (2) Regulation of pH
 - (3) Maintain level of body fluids
 - (4) All of these
- 157.** Which of the following is responsible for micturition?
- (1) Parasympathetic nerves
 - (2) Sympathetic nerves
 - (3) Both of these
 - (4) None of these
- 158.** GFR in a healthy individual is approximately
- (1) 125 ml/min
 - (2) 180 Litres/day
 - (3) 90 Litres/day
 - (4) Both (1) and (2)
- 159.** The renal threshold for glucose is
- (1) 100 mg/dL
 - (2) 120 mg/dL
 - (3) 180 mg/dL
 - (4) 250 mg/dL
- 160.** Accumulation of urea in blood of patients suffering with renal failure is called
- (1) Renal calculi
 - (2) Glomerulonephritis
 - (3) Uremia
 - (4) Ketosis

- 161.** Presence of pus in urine is called
 (1) Pyuria (2) Proteinuria
 (3) Albuminuria (4) Glycosuria
- 162.** Sebaceous glands secrete sebum containing
 (1) Waxes (2) Sterols
 (3) Fatty acids (4) All of these
- 163.** Scanty urine production is called
 (1) Polyuria (2) Oliguria
 (3) Pyuria (4) Haematuria
- 164.** For haemodialysis, blood is taken out of the body through:
 (1) Artery (2) Vein
 (3) Capillaries (4) Both (1) and (2)
- 165.** Which of the following is removed during haemodialysis?
 (1) Glucose (2) Amino acids
 (3) Urea (4) All of these
- 166.** Which of the following is not a character of skeletal muscle?
 (1) Syncytial (2) Striated
 (3) Voluntary (4) Branched
- 167.** In skeletal muscles, each muscle fibre is enclosed within
 (1) Epimysium
 (2) Endomysium
 (3) Perimysium
 (4) Endoneurium
- 168.** The type of muscle present in the wall of intestine is
 (1) Skeletal (2) Cardiac
 (3) Smooth (4) Both (1) and (2)
- 169.** Sarcomere is
 (1) Functional unit of skeletal muscle
 (2) Length between two Z-lines
 (3) Both (1) and (2)
 (4) Found in smooth muscle
- 170.** _____ is the monomer unit of myosin.
 (1) Meromyosin (2) Troponin
 (3) Tropomyosin (4) Actin
- 171.** Skeletal muscles
 (1) Are used in iris and ciliary body
 (2) Are present in myocardium
 (3) Are involuntary
 (4) Are attached with bones and used for locomotion
- 172.** Which of the following is incorrect regarding skeletal muscle?
 (1) Muscle fibre is multinucleated and called syncytial
 (2) Each myofibril has alternate dark and light bands
 (3) Dark bands contain mainly actin while light bands contain myosin
 (4) Sarcoplasmic reticulum is the store house of calcium
- 173.** All of the following statements are true regarding red muscle fibres, except
 (1) Fatigue occurs slowly
 (2) Rich Blood supply
 (3) Rich in mitochondria
 (4) Fast rate of contraction for short duration
- 174.** The entire muscle is covered by a sheath of connective tissue called
 (1) Endomysium (2) Peimysium
 (3) Epimysium (4) Epineurium
- 175.** During contraction of skeletal muscles, which of the following is observed?
 (1) Sliding of myosin filament over actin filament.
 (2) Sliding of actin filament over myosin filament
 (3) Myosin filaments coming close to each other
 (4) All of the above
- 176.** Which of the following is correct regarding white muscle fibres?
 (1) They are called aerobic muscles
 (2) They have rich blood supply
 (3) They are rich in mitochondria
 (4) Sarcoplasmic reticulum is abundant

177. The ion responsible for unmasking of active sites for myosin for cross bridge activity during muscle contraction

- (1) Calcium (2) Magnesium
(3) Sodium (4) Potassium

178. Myosin head has binding sites for

- (1) Actin and troponin
(2) Actin and ATP
(3) Troponin and tropomyosin
(4) Actin and tropomyosin

179. Which of the following has ATPase action required for muscle contraction?

- (1) Actin (2) Myosin
(3) Troponin (4) Tropomyosin

180. Which of the following is incorrect regarding protein filaments found in skeletal muscles?

- (1) 'G' (Globular) actin polymerize to form F-actin
(2) In meromyosin, globular head with short arm form heavy meromyosin (HMM) where as the tail of meromyosin forms light meromyosin (LMM)
(3) Thin filaments have actin, troponin and trimers of tropomyosin
(4) Myosin head has ATPase action which breaks down ATP to provide energy for muscle contraction



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